`TLS_ECDHE_PSK_WITH_CHACHA20_POLY1305_SHA256`	by ChaCha20-Poly1305.

All of the CCM cipher suites also come in a CCM_8 variant that uses a short 64-bit authentication tag. As previously discussed, these variants should only be used if you need to save every byte of network use and you are confident that you have alternative measures in place to ensure authenticity of network traffic. AES-GCM is also supported by PSK cipher suites, but I would not recommend it in constrained environments due to the increased risk of accidental nonce reuse.

12.3 End-to-end security

TLS and DTLS provide excellent security when an API client can talk directly to the server. But as mentioned in the introduction to section 12.1, in a typical IoT application messages may travel over multiple different protocols. For example, sensor data produced by devices may be sent over low-power wireless networks to a local gateway, which then puts them onto a MQTT message queue for transmission to another service, which aggregates the data and performs an HTTP POST request to a cloud REST API for analysis and storage. Although each hop on this journey can be secured using TLS, messages are available unencrypted while being processed at the intermediate nodes. This makes these intermediate nodes an attractive target for attackers, because once compromised they can view and manipulate all data flowing through that device.

The solution is to provide end-to-end security of all data independent of the transport layer security. Rather than